

The upper bound on the uniform separation rate $\rho(\Delta_\alpha^{\lambda,B}, \mathcal{S}_d^s(R), \beta, M)$ given by Theorem 6 consists of two terms depending on the bandwidth $\lambda \in (0, \infty)^d$. As the bandwidth λ varies, there is a trade-off between those two quantities: increasing one implies decreasing the other. We can choose the optimal bandwidth λ (depending on $m+n$, d and s) in the sense that both terms have the same order with respect to the sum of sample sizes $m+n$.

Corollary 7 (proof in Appendix E.7) *We assume that $\alpha \in (0, e^{-1})$, $\beta \in (0, 1)$, $s > 0$, $R > 0$, $M > 0$ and $B \in \mathbb{N}$ satisfying $B \geq \frac{3}{\alpha^2}(\ln(\frac{8}{\beta}) + \alpha(1-\alpha))$. The test $\Delta_\alpha^{\lambda^*,B}$ for the choice of bandwidth $\lambda_i^* = (m+n)^{-2/(4s+d)}$, $i = 1, \dots, d$, is optimal in the minimax sense over the Sobolev ball $\mathcal{S}_d^s(R)$, that is*

$$\rho(\Delta_\alpha^{\lambda^*,B}, \mathcal{S}_d^s(R), \beta, M) \leq C_5(M, d, s, R, \alpha, \beta) (m+n)^{-2s/(4s+d)}$$

for some positive constant $C_5(M, d, s, R, \alpha, \beta)$.

We have constructed the single MMD test $\Delta_\alpha^{\lambda^*,B}$ and proved that it is minimax optimal over the Sobolev ball $\mathcal{S}_d^s(R)$ without any restriction on the sample sizes m and n . However, it is worth pointing out that the optimality of the single MMD test hinges on the assumption that the smoothness parameter s is known in advance, which is not realistic. Given this limitation, our next goal is to construct a test which does not rely on the unknown smoothness parameter s of the Sobolev ball $\mathcal{S}_d^s(R)$ and achieves the same minimax rate, up to an iterated logarithmic term, for all $s > 0$ and $R > 0$. This is the main topic of Section 3.5 below.

3.5 Non-asymptotic MMDAgg test aggregating multiple bandwidths

We propose to construct an aggregated test (MMDAgg) by combining multiple single MMD tests, which allows the test to be adaptive to the unknown the smoothness parameter of the Sobolev balls. We use the powerful multiple testing correction of Romano and Wolf (2005b, Equation 9), for which we derive non-asymptotic level guarantees. Consider a finite collection Λ of bandwidths in $(0, \infty)^d$ with an associated collection of positive weights³ $(w_\lambda)_{\lambda \in \Lambda}$, which will determine the importance of each single MMD test over the others when aggregating all of them. We require that $\sum_{\lambda \in \Lambda} w_\lambda \leq 1$. For notational convenience, we let Λ^w denote the collection of bandwidths Λ with its associated collection of weights. Intuitively, we want to define our aggregated MMDAgg test as the test which rejects the null hypothesis $\mathcal{H}_0: p = q$ if one of the single MMD tests $(\Delta_{u_\alpha w_\lambda}^{\lambda, B_1})_{\lambda \in \Lambda}$ rejects the null hypothesis, where u_α is defined as⁴

$$u_\alpha = \sup \left\{ u \in \left(0, \min_{\lambda \in \Lambda} w_\lambda^{-1} \right) : \mathbb{P}_{p \times p \times r} \left(\max_{\lambda \in \Lambda} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) - \widehat{q}_{1-u w_\lambda}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) > 0 \right) \leq \alpha \right\}$$

3. We stress that this differs from the notation often used in the literature (for example, for the independence aggregated test of Albert et al., 2022) where the weights are defined as e^{-w_λ} rather than as w_λ .

4. Since $\alpha \in (0, 1)$ and the function $u \mapsto \mathbb{P}_{p \times p \times r}(\max_{\lambda \in \Lambda}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) - \widehat{q}_{1-u w_\lambda}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n)) > 0)$ is non-decreasing, tends to 0 as u tends to 0, and tends to 1 as u tends to $\min_{\lambda \in \Lambda} w_\lambda^{-1}$, u_α is well-defined.

to ensure that MMDAgg has level α . We stress that the data, as well as the choice of collections of bandwidths and weights, all affect the value of u_α . In practice, the probability and the supremum in the definition of u_α cannot be computed exactly. We can estimate the former using a Monte Carlo approximation and estimate the latter using the bisection method. We now explain this in more detail and provide a formal definition of our aggregated MMDAgg test.

For the case of the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ defined in Equation (3), we independently generate a permutation $\sigma^{(b, \ell)} \sim r$ of $\{1, \dots, m+n\}$ and compute $\widehat{M}_{\lambda, \ell}^b := \widehat{M}_\lambda^{\sigma^{(b, \ell)}}$ as defined in Equation (10) for $\ell = 1, 2$, $b = 1, \dots, B_\ell$ and $\lambda \in \Lambda$. When working with the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ defined in Equation (6), we independently generate n i.i.d. Rademacher random variables $\epsilon^{(b, \ell)} = (\epsilon_1^{(b, \ell)}, \dots, \epsilon_n^{(b, \ell)}) \sim r$ and compute $\widehat{M}_{\lambda, \ell}^b := \widehat{M}_\lambda^{\epsilon^{(b, \ell)}}$ as defined in Equation (11) for $\ell = 1, 2$, $b = 1, \dots, B_\ell$ and $\lambda \in \Lambda$. For consistency between the two procedures, we let $\mathbb{Z}_{B_\ell}^\ell := (\mu^{(b, \ell)})_{1 \leq b \leq B_\ell}$ for $\ell = 1, 2$, where $\mu^{(b, \ell)}$ denotes either the permutation $\sigma^{(b, \ell)}$ or the Rademacher random variable $\epsilon^{(b, \ell)}$ for $\ell = 1, 2$ and $b = 1, \dots, B_\ell$. With a slight abuse of notation, we refer to $\mathbb{Z}_{B_1}^1$ and $\mathbb{Z}_{B_2}^2$ simply as $\mathbb{Z}_{B_1}$ and $\mathbb{Z}_{B_2}$. For both estimators, we also let $\widehat{M}_{\lambda, 1}^{B_1+1} := \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$. We denote by $\widehat{M}_{\lambda, 1}^{\bullet 1} \leq \dots \leq \widehat{M}_{\lambda, 1}^{\bullet B_1+1}$ the ordered elements $(\widehat{M}_{\lambda, 1}^b)_{1 \leq b \leq B_1+1}$.

We use $(\widehat{M}_{\lambda, 1}^{\bullet b})_{1 \leq b \leq B_1+1}$, which are computed using $\mathbb{Z}_{B_1}$, $\mathbb{X}_m$ and $\mathbb{Y}_n$, to estimate the conditional $(1-a)$ -quantile

$$\widehat{q}_{1-a}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) := \widehat{M}_{\lambda, 1}^{\bullet \lceil (B_1+1)(1-a) \rceil}$$

for any $a \in (0, 1)$ as in Equation (12). As explained in Section 3.2.3, $\widehat{q}_{1-a}^{\lambda, B_1}$ is defined such that an a -proportion of the test statistics $(\widehat{M}_{\lambda, 1}^b)_{1 \leq b \leq B_1+1}$ simulated under the null are greater than $\widehat{q}_{1-a}^{\lambda, B_1}$. By Proposition 1, this ensures the single test with bandwidth λ has non-asymptotic level a .

We use $(\widehat{M}_{\lambda, 2}^{\bullet b})_{1 \leq b \leq B_2}$, which are computed using $\mathbb{Z}_{B_2}$, $\mathbb{X}_m$ and $\mathbb{Y}_n$, to estimate with a Monte Carlo approximation the probability

$$\mathbb{P}_{p \times q \times r} \left(\max_{\lambda \in \Lambda} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) - \widehat{q}_{1-uw\lambda}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) > 0 \right) \quad (13)$$

which appears in the definition of u_α . We denote the approximated quantity by $u_\alpha^{B_2}$, which is formally defined as

$$\begin{aligned} & u_\alpha^{B_2}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) \\ &:= \sup \left\{ u \in (0, \min_{\lambda \in \Lambda} w_\lambda^{-1}) : \frac{1}{B_2} \sum_{b=1}^{B_2} \mathbb{1} \left(\max_{\lambda \in \Lambda} \left(\widehat{M}_{\lambda, 2}^b(\mu^{(b, 2)} | \mathbb{X}_m, \mathbb{Y}_n) - \widehat{q}_{1-uw\lambda}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) > 0 \right) \leq \alpha \right\} \\ &= \sup \left\{ u \in (0, \min_{\lambda \in \Lambda} w_\lambda^{-1}) : \frac{1}{B_2} \sum_{b=1}^{B_2} \mathbb{1} \left(\max_{\lambda \in \Lambda} \left(\widehat{M}_{\lambda, 2}^b - \widehat{M}_{\lambda, 1}^{\bullet \lceil (B_1+1)(1-uw\lambda) \rceil} \right) > 0 \right) \leq \alpha \right\}. \end{aligned}$$

Since the function $u \mapsto \frac{1}{B_2} \sum_{b=1}^{B_2} \mathbb{1} \left(\max_{\lambda \in \Lambda} \left(\widehat{M}_{\lambda, 2}^b - \widehat{M}_{\lambda, 1}^{\bullet \lceil (B_1+1)(1-uw\lambda) \rceil} \right) > 0 \right) - \alpha$ is increasing, $u_\alpha^{B_2}$ is actually the largest root of this function. As such, it can be computed in practice by

using the bisection method for finding the root. We let⁵ $\hat{u}_\alpha^{B_{2:3}} = \hat{u}_\alpha^{B_{2:3}}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1})$ be the lower bound of the interval obtained by performing B_3 steps of the bisection method to approximate the supremum (*i.e.* find the root) in the definition of $u_\alpha^{B_2}$. We then have

$$u_\alpha^{B_2} \in \left[\hat{u}_\alpha^{B_{2:3}}, \hat{u}_\alpha^{B_{2:3}} + 2^{-B_3} \min_{\lambda \in \Lambda} w_\lambda^{-1} \right].$$

We recall that the data, the collection of bandwidths, and the weights, all affect the value of the correction u_α , and hence, also the value of its estimate $\hat{u}_\alpha^{B_{2:3}}$.

For $\alpha \in (0, 1)$, we can then define our aggregated MMDAgg test⁵ $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ as rejecting the null hypothesis, that is $\Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}) = 1$, if one of the tests $\left(\Delta_{\hat{u}_\alpha^{B_{2:3}} w_\lambda}^{\lambda, B_1} \right)_{\lambda \in \Lambda}$ rejects the null hypothesis, that is

$$\exists \lambda \in \Lambda : \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) > \hat{q}_{1 - \hat{u}_\alpha^{B_{2:3}}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) w_\lambda}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n),$$

or equivalently

$$\exists \lambda \in \Lambda : \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) > \widehat{M}_{\lambda, 1}^{\bullet, (B_1+1)(1 - \hat{u}_\alpha^{B_{2:3}}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) w_\lambda)}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n).$$

The parameters of our MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ are: its level α , the finite collection Λ^w of bandwidths with its associated weights, and the positive integers B_1 , B_2 and B_3 . We generate independent permutations or Rademacher random variables to obtain $\mathbb{Z}_{B_1}$ and $\mathbb{Z}_{B_2}$. In practice, we are given realisations of $\mathbb{X}_m = (X_i)_{1 \leq i \leq m}$ and $\mathbb{Y}_n = (Y_j)_{1 \leq j \leq n}$. Hence, we are able to compute $\Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2})$ to decide whether or not we should reject the null hypothesis $\mathcal{H}_0: p = q$. This exact version of our aggregated MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ can be implemented in practice with no further approximation. We provide a detailed pseudocode of MMDAgg in Algorithm 1 and our code is available here. In Appendix C, we further discuss how to efficiently compute the values $\widehat{M}_{\lambda, \ell}^b$ for $\ell = 1, 2$, $b = 1, \dots, B_\ell$ and $\lambda \in \Lambda$ (corresponding to Step 1 of Algorithm 1).

The only conditions we have on our weights $(w_\lambda)_{\lambda \in \Lambda}$ for the collection of bandwidths Λ are that they need to be positive and to satisfy $\sum_{\lambda \in \Lambda} w_\lambda \leq 1$. We now explain why this condition on the sum of the weights is not necessarily required. In general, the two aggregated tests with weights $(w_\lambda)_{\lambda \in \Lambda}$ and with scaled weights $(w'_\lambda)_{\lambda \in \Lambda}$ where $w'_\lambda := \frac{w_\lambda}{\sum_{\lambda \in \Lambda} w_\lambda}$ for $\lambda \in \Lambda$ are exactly the same. This is due to the way the correction of the levels of the single MMD tests is performed. In particular, making the dependence of $\hat{u}_\alpha^{B_{2:3}}$ on either Λ^w or $\Lambda^{w'}$ explicit, we have

$$\hat{u}_\alpha^{B_{2:3}, \Lambda^{w'}}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) = \hat{u}_\alpha^{B_{2:3}, \Lambda^w}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) \sum_{\lambda \in \Lambda} w_\lambda,$$

and so $\hat{u}_\alpha^{B_{2:3}, \Lambda^{w'}} w'_\lambda = \hat{u}_\alpha^{B_{2:3}, \Lambda^w} w_\lambda$, which implies that

$$\Delta_\alpha^{\Lambda^{w'}, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}) = \Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}). \quad (14)$$

5. We use the condensed notation $B_{2:3}$ and $B_{1:3}$ to refer to (B_2, B_3) and (B_1, B_2, B_3) , respectively.